

FIGURE 7

Note, too, that the equation

$$(7) \quad z = Re^{i\theta} \quad (0 \leq \theta \leq 2\pi)$$

is a parametric representation of the circle $|z| = R$, centered at the origin with radius R . As the parameter θ increases from $\theta = 0$ to $\theta = 2\pi$, the point z starts from the positive real axis and traverses the circle once in the counterclockwise direction. More generally, the circle $|z - z_0| = R$, whose center is z_0 and whose radius is R , has the parametric representation

$$(8) \quad z = z_0 + Re^{i\theta} \quad (0 \leq \theta \leq 2\pi).$$

This can be seen vectorially (Fig. 8) by noting that a point z traversing the circle $|z - z_0| = R$ once in the counterclockwise direction corresponds to the sum of the fixed vector z_0 and a vector of length R whose angle of inclination θ varies from $\theta = 0$ to $\theta = 2\pi$.

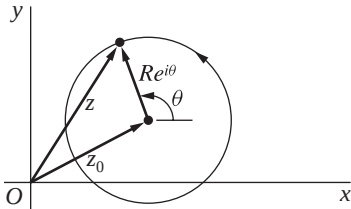


FIGURE 8

7. PRODUCTS AND POWERS IN EXPONENTIAL FORM

Simple trigonometry tells us that $e^{i\theta}$ has the familiar additive property of the exponential function in calculus:

$$\begin{aligned} e^{i\theta_1} e^{i\theta_2} &= (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \\ &= \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) = e^{i(\theta_1 + \theta_2)}. \end{aligned}$$

Thus, if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, the product $z_1 z_2$ has exponential form

$$(1) \quad z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = (r_1 r_2) e^{i(\theta_1 + \theta_2)}.$$

Furthermore,

$$(2) \quad \frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} \cdot \frac{e^{i\theta_1} e^{-i\theta_2}}{e^{i\theta_2} e^{-i\theta_2}} = \frac{r_1}{r_2} \cdot \frac{e^{i(\theta_1 - \theta_2)}}{e^{i0}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Note how it follows from expression (2) that the inverse of any nonzero complex number $z = r e^{i\theta}$ is

$$(3) \quad z^{-1} = \frac{1}{z} = \frac{1 e^{i0}}{r e^{i\theta}} = \frac{1}{r} e^{i(0 - \theta)} = \frac{1}{r} e^{-i\theta}.$$

Expressions (1), (2), and (3) are, of course, easily remembered by applying the usual algebraic rules for real numbers and e^x .

Another important result that can be obtained formally by applying rules for real numbers to $z = r e^{i\theta}$ is

$$(4) \quad z^n = r^n e^{in\theta} \quad (n = 0, \pm 1, \pm 2, \dots).$$

It is easily verified for positive values of n by mathematical induction. To be specific, we first note that it becomes $z = r e^{i\theta}$ when $n = 1$. Next, we assume that it is valid when $n = m$, where m is any positive integer. In view of expression (1) for the product of two nonzero complex numbers in exponential form, it is then valid for $n = m + 1$:

$$z^{m+1} = z^m z = r^m e^{im\theta} r e^{i\theta} = (r^m r) e^{i(m\theta + \theta)} = r^{m+1} e^{i(m+1)\theta}.$$

Expression (4) is thus verified when n is a positive integer. It also holds when $n = 0$, with the convention that $z^0 = 1$. If $n = -1, -2, \dots$, on the other hand, we define z^n in terms of the multiplicative inverse of z by writing

$$z^n = (z^{-1})^m \quad \text{where} \quad m = -n = 1, 2, \dots$$

Then, since equation (4) is valid for positive integers, it follows from the exponential form (3) of z^{-1} that

$$z^n = \left[\frac{1}{r} e^{i(-\theta)} \right]^m = \left(\frac{1}{r} \right)^m e^{im(-\theta)} = \left(\frac{1}{r} \right)^{-n} e^{i(-n)(-\theta)} = r^n e^{in\theta} \quad (n = -1, -2, \dots).$$

Expression (4) is now established for all integral powers.

Expression (4) can be useful in finding powers of complex numbers even when they are given in rectangular form and the result is desired in that form.

EXAMPLE 1. In order to put $(\sqrt{3} + i)^7$ in rectangular form, one need only write

$$(\sqrt{3} + i)^7 = (2e^{i\pi/6})^7 = 2^7 e^{i7\pi/6} = (2^6 e^{i\pi})(2e^{i\pi/6}) = -64(\sqrt{3} + i).$$

Finally, we observe that if $r = 1$, equation (4) becomes

$$(5) \quad (e^{i\theta})^n = e^{in\theta} \quad (n = 0, \pm 1, \pm 2, \dots).$$

When written in the form

$$(6) \quad (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (n = 0, \pm 1, \pm 2, \dots),$$

this is known as *de Moivre's formula*. The following example uses a special case of it.

EXAMPLE 2. Formula (6) with $n = 2$ tells us that

$$(\cos \theta + i \sin \theta)^2 = \cos 2\theta + i \sin 2\theta,$$

or

$$\cos^2 \theta - \sin^2 \theta + i 2 \sin \theta \cos \theta = \cos 2\theta + i \sin 2\theta.$$

By equating real parts and then imaginary parts here, we have the familiar trigonometric identities

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta.$$

(See also Exercises 10 and 11, Sec. 8.)

8. ARGUMENTS OF PRODUCTS AND QUOTIENTS

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, the expression

$$(1) \quad z_1 z_2 = (r_1 r_2) e^{i(\theta_1 + \theta_2)}$$

in Sec. 7 can be used to obtain an important identity involving arguments:

$$(2) \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2.$$

This result is to be interpreted as saying that if values of two of the three (multiple-valued) arguments are specified, then there is a value of the third such that the equation holds.

We start the verification of statement (2) by letting θ_1 and θ_2 denote any values of $\arg z_1$ and $\arg z_2$, respectively. Expression (1) then tells us that $\theta_1 + \theta_2$ is a value of $\arg(z_1 z_2)$. (See Fig. 9.) If, on the other hand, values of $\arg(z_1 z_2)$ and

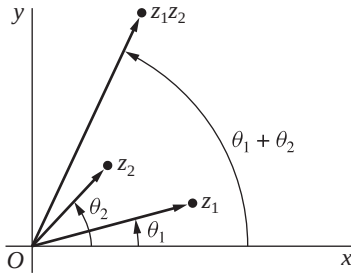


FIGURE 9

$\arg z_1$ are specified, those values correspond to particular choices of n and n_1 in the expressions

$$\arg(z_1 z_2) = (\theta_1 + \theta_2) + 2n\pi \quad (n = 0, \pm 1, \pm 2, \dots)$$

and

$$\arg z_1 = \theta_1 + 2n_1\pi \quad (n_1 = 0, \pm 1, \pm 2, \dots).$$

Since

$$(\theta_1 + \theta_2) + 2n\pi = (\theta_1 + 2n_1\pi) + [\theta_2 + 2(n - n_1)\pi],$$

equation (2) is evidently satisfied when the value

$$\arg z_2 = \theta_2 + 2(n - n_1)\pi$$

is chosen. Verification when values of $\arg(z_1 z_2)$ and $\arg z_2$ are specified follows by symmetry.

Statement (2) is sometimes valid when $\arg$ is replaced everywhere by Arg (see Exercise 6). But, as the following example illustrates, that is *not always* the case.

EXAMPLE 1. When $z_1 = -1$ and $z_2 = i$,

$$\text{Arg}(z_1 z_2) = \text{Arg}(-i) = -\frac{\pi}{2} \quad \text{but} \quad \text{Arg } z_1 + \text{Arg } z_2 = \pi + \frac{\pi}{2} = \frac{3\pi}{2}.$$

If, however, we take the values of $\arg z_1$ and $\arg z_2$ just used and select the value

$$\text{Arg}(z_1 z_2) + 2\pi = -\frac{\pi}{2} + 2\pi = \frac{3\pi}{2}$$

of $\arg(z_1 z_2)$, we find that equation (2) is satisfied.

Statement (2) tells us that

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1 z_2^{-1}) = \arg z_1 + \arg(z_2^{-1});$$

and, since (Sec. 7)

$$z_2^{-1} = \frac{1}{r_2} e^{-i\theta_2},$$

one can see that

$$(3) \quad \arg(z_2^{-1}) = -\arg z_2.$$

Hence

$$(4) \quad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2.$$

Statement (3) is, of course, to be interpreted as saying that the set of all values on the left-hand side is the same as the set of all values on the right-hand side. Statement (4) is, then, to be interpreted in the same way that statement (2) is.

EXAMPLE 2. In order to find the principal argument $\text{Arg } z$ when

$$z = \frac{-2}{1 + \sqrt{3}i},$$

observe that

$$\arg z = \arg(-2) - \arg(1 + \sqrt{3}i).$$

Since

$$\text{Arg}(-2) = \pi \quad \text{and} \quad \text{Arg}(1 + \sqrt{3}i) = \frac{\pi}{3},$$

one value of $\arg z$ is $2\pi/3$; and, because $2\pi/3$ is between $-\pi$ and π , we find that $\text{Arg } z = 2\pi/3$.

EXERCISES

1. Find the principal argument $\text{Arg } z$ when

$$(a) \ z = \frac{i}{-2 - 2i}; \quad (b) \ z = (\sqrt{3} - i)^6.$$

$$\text{Ans. } (a) \ -3\pi/4; \quad (b) \ \pi.$$

2. Show that (a) $|e^{i\theta}| = 1$; (b) $\overline{e^{i\theta}} = e^{-i\theta}$.

3. Use mathematical induction to show that

$$e^{i\theta_1} e^{i\theta_2} \dots e^{i\theta_n} = e^{i(\theta_1 + \theta_2 + \dots + \theta_n)} \quad (n = 2, 3, \dots).$$

4. Using the fact that the modulus $|e^{i\theta} - 1|$ is the distance between the points $e^{i\theta}$ and 1 (see Sec. 4), give a geometric argument to find a value of θ in the interval $0 \leq \theta < 2\pi$ that satisfies the equation $|e^{i\theta} - 1| = 2$.

$$\text{Ans. } \pi.$$

5. By writing the individual factors on the left in exponential form, performing the needed operations, and finally changing back to rectangular coordinates, show that

$$(a) i(1 - \sqrt{3}i)(\sqrt{3} + i) = 2(1 + \sqrt{3}i); \quad (b) 5i/(2 + i) = 1 + 2i;$$

$$(c) (-1 + i)^7 = -8(1 + i); \quad (d) (1 + \sqrt{3}i)^{-10} = 2^{-11}(-1 + \sqrt{3}i).$$

6. Show that if $\operatorname{Re} z_1 > 0$ and $\operatorname{Re} z_2 > 0$, then

$$\operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2,$$

where principal arguments are used.

7. Let z be a nonzero complex number and n a negative integer ($n = -1, -2, \dots$). Also, write $z = re^{i\theta}$ and $m = -n = 1, 2, \dots$. Using the expressions

$$z^m = r^m e^{im\theta} \quad \text{and} \quad z^{-1} = \left(\frac{1}{r}\right) e^{i(-\theta)},$$

verify that $(z^m)^{-1} = (z^{-1})^m$ and hence that the definition $z^n = (z^{-1})^m$ in Sec. 7 could have been written alternatively as $z^n = (z^m)^{-1}$.

8. Prove that two nonzero complex numbers z_1 and z_2 have the same moduli if and only if there are complex numbers c_1 and c_2 such that $z_1 = c_1 c_2$ and $z_2 = c_1 \overline{c_2}$.

Suggestion: Note that

$$\exp\left(i\frac{\theta_1 + \theta_2}{2}\right) \exp\left(i\frac{\theta_1 - \theta_2}{2}\right) = \exp(i\theta_1)$$

and [see Exercise 2(b)]

$$\exp\left(i\frac{\theta_1 + \theta_2}{2}\right) \overline{\exp\left(i\frac{\theta_1 - \theta_2}{2}\right)} = \exp(i\theta_2).$$

9. Establish the identity

$$1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z} \quad (z \neq 1)$$

and then use it to derive *Lagrange's trigonometric identity*:

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin[(2n+1)\theta/2]}{2 \sin(\theta/2)} \quad (0 < \theta < 2\pi).$$

Suggestion: As for the first identity, write $S = 1 + z + z^2 + \dots + z^n$ and consider the difference $S - zS$. To derive the second identity, write $z = e^{i\theta}$ in the first one.

10. Use de Moivre's formula (Sec. 7) to derive the following trigonometric identities:

$$(a) \cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta; \quad (b) \sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta.$$

11. (a) Use the binomial formula (Sec. 3) and de Moivre's formula (Sec. 7) to write

$$\cos n\theta + i \sin n\theta = \sum_{k=0}^n \binom{n}{k} \cos^{n-k} \theta (i \sin \theta)^k \quad (n = 0, 1, 2, \dots).$$

Then define the integer m by means of the equations

$$m = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ (n-1)/2 & \text{if } n \text{ is odd} \end{cases}$$

and use the above summation to show that [compare with Exercise 10(a)]

$$\cos n\theta = \sum_{k=0}^m \binom{n}{2k} (-1)^k \cos^{n-2k} \theta \sin^{2k} \theta \quad (n = 0, 1, 2, \dots).$$

(b) Write $x = \cos \theta$ in the final summation in part (a) to show that it becomes a polynomial

$$T_n(x) = \sum_{k=0}^m \binom{n}{2k} (-1)^k x^{n-2k} (1-x^2)^k$$

of degree n ($n = 0, 1, 2, \dots$) in the variable x .*

9. ROOTS OF COMPLEX NUMBERS

Consider now a point $z = re^{i\theta}$, lying on a circle centered at the origin with radius r (Fig. 10). As θ is increased, z moves around the circle in the counterclockwise direction. In particular, when θ is increased by 2π , we arrive at the original point; and the same is true when θ is decreased by 2π . It is, therefore, evident from Fig. 10 that *two nonzero complex numbers*

$$z_1 = r_1 e^{i\theta_1} \quad \text{and} \quad z_2 = r_2 e^{i\theta_2}$$

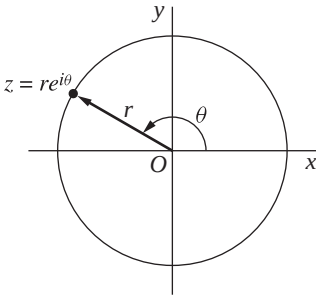


FIGURE 10

*These are called Chebyshev polynomials and are prominent in approximation theory.

are equal if and only if

$$r_1 = r_2 \quad \text{and} \quad \theta_1 = \theta_2 + 2k\pi,$$

where k is some integer ($k = 0, \pm 1, \pm 2, \dots$).

This observation, together with the expression $z^n = r^n e^{in\theta}$ in Sec. 7 for integral powers of complex numbers $z = re^{i\theta}$, is useful in finding the n th roots of any nonzero complex number $z_0 = r_0 e^{i\theta_0}$, where n has one of the values $n = 2, 3, \dots$. The method starts with the fact that an n th root of z_0 is a nonzero number $z = re^{i\theta}$ such that $z^n = z_0$, or

$$r^n e^{in\theta} = r_0 e^{i\theta_0}.$$

According to the statement in italics just above, then,

$$r^n = r_0 \quad \text{and} \quad n\theta = \theta_0 + 2k\pi,$$

where k is any integer ($k = 0, \pm 1, \pm 2, \dots$). So $r = \sqrt[n]{r_0}$, where this radical denotes the unique positive n th root of the positive real number r_0 , and

$$\theta = \frac{\theta_0 + 2k\pi}{n} = \frac{\theta_0}{n} + \frac{2k\pi}{n} \quad (k = 0, \pm 1, \pm 2, \dots).$$

Consequently, the complex numbers

$$z = \sqrt[n]{r_0} \exp \left[i \left(\frac{\theta_0}{n} + \frac{2k\pi}{n} \right) \right] \quad (k = 0, \pm 1, \pm 2, \dots)$$

are the n th roots of z_0 . We are able to see immediately from this exponential form of the roots that they all lie on the circle $|z| = \sqrt[n]{r_0}$ about the origin and are equally spaced every $2\pi/n$ radians, starting with argument θ_0/n . Evidently, then, all of the *distinct* roots are obtained when $k = 0, 1, 2, \dots, n-1$, and no further roots arise with other values of k . We let c_k ($k = 0, 1, 2, \dots, n-1$) denote these distinct roots and write

$$(1) \quad c_k = \sqrt[n]{r_0} \exp \left[i \left(\frac{\theta_0}{n} + \frac{2k\pi}{n} \right) \right] \quad (k = 0, 1, 2, \dots, n-1).$$

(See Fig. 11.)

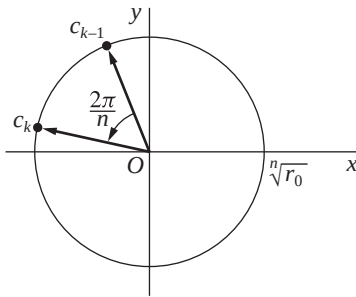


FIGURE 11

The number $\sqrt[n]{r_0}$ is the length of each of the radius vectors representing the n roots. The first root c_0 has argument θ_0/n ; and the two roots when $n = 2$ lie at the opposite ends of a diameter of the circle $|z| = \sqrt[n]{r_0}$, the second root being $-c_0$. When $n \geq 3$, the roots lie at the vertices of a regular polygon of n sides inscribed in that circle.

We shall let $z_0^{1/n}$ denote the set of n th roots of z_0 . If, in particular, z_0 is a positive real number r_0 , the symbol $r_0^{1/n}$ denotes the entire set of roots; and the symbol $\sqrt[n]{r_0}$ in expression (1) is reserved for the one positive root. When the value of θ_0 that is used in expression (1) is the principal value of $\arg z_0$ ($-\pi < \theta_0 \leq \pi$), the number c_0 is referred to as the *principal root*. Thus when z_0 is a positive real number r_0 , its principal root is $\sqrt[n]{r_0}$.

Observe that if we write expression (1) for the roots of z_0 as

$$c_k = \sqrt[n]{r_0} \exp\left(i \frac{\theta_0}{n}\right) \exp\left(i \frac{2k\pi}{n}\right) \quad (k = 0, 1, 2, \dots, n-1),$$

and also write

$$(2) \quad \omega_n = \exp\left(i \frac{2\pi}{n}\right),$$

it follows from property (5), Sec. 7. of $e^{i\theta}$ that

$$(3) \quad \omega_n^k = \exp\left(i \frac{2k\pi}{n}\right) \quad (k = 0, 1, 2, \dots, n-1)$$

and hence that

$$(4) \quad c_k = c_0 \omega_n^k \quad (k = 0, 1, 2, \dots, n-1).$$

The number c_0 here can, of course, be replaced by any particular n th root of z_0 , since ω_n represents a counterclockwise rotation through $2\pi/n$ radians.

Finally, a convenient way to remember expression (1) is to write z_0 in its most general exponential form (compare with Example 2 in Sec. 6)

$$(5) \quad z_0 = r_0 e^{i(\theta_0 + 2k\pi)} \quad (k = 0, \pm 1, \pm 2, \dots)$$

and to *formally* apply laws of fractional exponents involving real numbers, keeping in mind that there are precisely n roots:

$$z_0^{1/n} = [r_0 e^{i(\theta_0 + 2k\pi)}]^{1/n} = \sqrt[n]{r_0} \exp\left[\frac{i(\theta_0 + 2k\pi)}{n}\right] = \sqrt[n]{r_0} \exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right] \\ (k = 0, 1, 2, \dots, n-1).$$

The examples in the next section serve to illustrate this method for finding roots of complex numbers.

10. EXAMPLES

In each of the examples here, we start with expression (5), Sec. 9, and proceed in the manner described just after it.

EXAMPLE 1. Let us find all values of $(-8i)^{1/3}$, or the three cube roots of the number $-8i$. One need only write

$$-8i = 8 \exp\left[i\left(-\frac{\pi}{2} + 2k\pi\right)\right] \quad (k = 0, \pm 1, \pm 2, \dots)$$

to see that the desired roots are

$$(1) \quad c_k = 2 \exp\left[i\left(-\frac{\pi}{6} + \frac{2k\pi}{3}\right)\right] \quad (k = 0, 1, 2).$$

They lie at the vertices of an equilateral triangle, inscribed in the circle $|z| = 2$, and are equally spaced around that circle every $2\pi/3$ radians, starting with the principal root (Fig. 12)

$$c_0 = 2 \exp\left[i\left(-\frac{\pi}{6}\right)\right] = 2\left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right) = \sqrt{3} - i.$$

Without any further calculations, it is then evident that $c_1 = 2i$; and, since c_2 is symmetric to c_0 with respect to the imaginary axis, we know that $c_2 = -\sqrt{3} - i$.

Note how it follows from expressions (2) and (4) in Sec. 9 that these roots can be written

$$c_0, c_0\omega_3, c_0\omega_3^2 \quad \text{where} \quad \omega_3 = \exp\left(i\frac{2\pi}{3}\right).$$

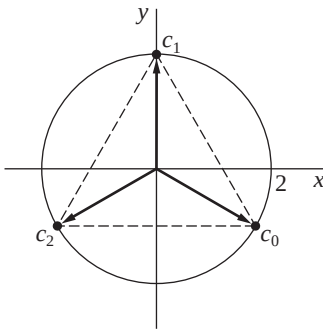


FIGURE 12

EXAMPLE 2. In order to determine the n th roots of unity, we start with

$$1 = 1 \exp[i(0 + 2k\pi)] \quad (k = 0, \pm 1, \pm 2 \dots)$$

and find that

$$(2) \quad 1^{1/n} = \sqrt[n]{1} \exp\left[i\left(\frac{0}{n} + \frac{2k\pi}{n}\right)\right] = \exp\left(i\frac{2k\pi}{n}\right) \quad (k = 0, 1, 2, \dots, n-1).$$

When $n = 2$, these roots are, of course, ± 1 . When $n \geq 3$, the regular polygon at whose vertices the roots lie is inscribed in the unit circle $|z| = 1$, with one vertex corresponding to the principal root $z = 1$ ($k = 0$). In view of expression (3), Sec. 9, these roots are simply

$$1, \omega_n, \omega_n^2, \dots, \omega_n^{n-1} \quad \text{where} \quad \omega_n = \exp\left(i\frac{2\pi}{n}\right).$$

See Fig. 13, where the cases $n = 3, 4$, and 6 are illustrated. Note that $\omega_n^n = 1$.

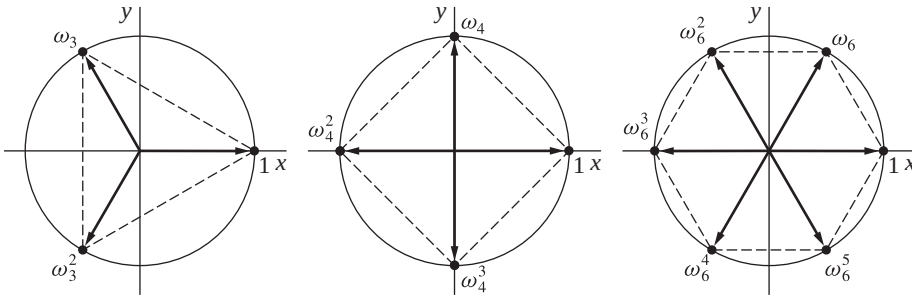


FIGURE 13

EXAMPLE 3. The two values c_k ($k = 0, 1$) of $(\sqrt{3} + i)^{1/2}$, which are the square roots of $\sqrt{3} + i$, are found by writing

$$\sqrt{3} + i = 2 \exp\left[i\left(\frac{\pi}{6} + 2k\pi\right)\right] \quad (k = 0, \pm 1, \pm 2, \dots)$$

and (see Fig. 14)

$$(3) \quad c_k = \sqrt{2} \exp\left[i\left(\frac{\pi}{12} + k\pi\right)\right] \quad (k = 0, 1).$$

Euler's formula tells us that

$$c_0 = \sqrt{2} \exp\left(i\frac{\pi}{12}\right) = \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right),$$

and the trigonometric identities

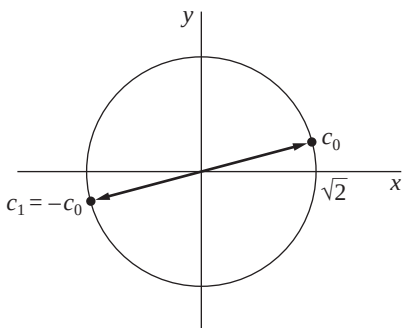


FIGURE 14

$$(4) \quad \cos^2 \frac{\alpha}{2} = \frac{1 + \cos \alpha}{2}, \quad \sin^2 \frac{\alpha}{2} = \frac{1 - \cos \alpha}{2}$$

enable us to write

$$\cos^2 \frac{\pi}{12} = \frac{1}{2} \left(1 + \cos \frac{\pi}{6} \right) = \frac{1}{2} \left(1 + \frac{\sqrt{3}}{2} \right) = \frac{2 + \sqrt{3}}{4},$$

$$\sin^2 \frac{\pi}{12} = \frac{1}{2} \left(1 - \cos \frac{\pi}{6} \right) = \frac{1}{2} \left(1 - \frac{\sqrt{3}}{2} \right) = \frac{2 - \sqrt{3}}{4}.$$

Consequently,

$$c_0 = \sqrt{2} \left(\sqrt{\frac{2 + \sqrt{3}}{4}} + i \sqrt{\frac{2 - \sqrt{3}}{4}} \right) = \frac{1}{\sqrt{2}} \left(\sqrt{2 + \sqrt{3}} + i \sqrt{2 - \sqrt{3}} \right).$$

Since $c_1 = -c_0$, the two square roots of $\sqrt{3} + i$ are, then,

$$(5) \quad \pm \frac{1}{\sqrt{2}} \left(\sqrt{2 + \sqrt{3}} + i \sqrt{2 - \sqrt{3}} \right).$$

EXERCISES

- Find the square roots of (a) $2i$; (b) $1 - \sqrt{3}i$ and express them in rectangular coordinates.

$$\text{Ans. (a) } \pm (1 + i); \quad (b) \pm \frac{\sqrt{3} - i}{\sqrt{2}}.$$

- In each case, find all the roots in rectangular coordinates, exhibit them as vertices of certain squares, and point out which is the principal root:

$$(a) (-16)^{1/4}; \quad (b) (-8 - 8\sqrt{3}i)^{1/4}.$$

$$\text{Ans. (a) } \pm \sqrt{2}(1 + i), \pm \sqrt{2}(1 - i); \quad (b) \pm(\sqrt{3} - i), \pm(1 + \sqrt{3}i).$$

3. In each case, find all the roots in rectangular coordinates, exhibit them as vertices of certain regular polygons, and identify the principal root:

(a) $(-1)^{1/3}$; (b) $8^{1/6}$.

Ans. (b) $\pm\sqrt{2}$, $\pm\frac{1+\sqrt{3}i}{\sqrt{2}}$, $\pm\frac{1-\sqrt{3}i}{\sqrt{2}}$.

4. According to Sec. 9, the three cube roots of a nonzero complex number z_0 can be written $c_0, c_0\omega_3, c_0\omega_3^2$ where c_0 is the principal cube root of z_0 and

$$\omega_3 = \exp\left(i\frac{2\pi}{3}\right) = \frac{-1 + \sqrt{3}i}{2}.$$

Show that if $z_0 = -4\sqrt{2} + 4\sqrt{2}i$, then $c_0 = \sqrt{2}(1 + i)$ and the other two cube roots are, in rectangular form, the numbers

$$c_0\omega_3 = \frac{-(\sqrt{3}+1) + (\sqrt{3}-1)i}{\sqrt{2}}, \quad c_0\omega_3^2 = \frac{(\sqrt{3}-1) - (\sqrt{3}+1)i}{\sqrt{2}}.$$

5. (a) Let a denote any fixed real number and show that the two square roots of $a + i$ are

$$\pm\sqrt{A}\exp\left(i\frac{\alpha}{2}\right)$$

where $A = \sqrt{a^2 + 1}$ and $\alpha = \text{Arg}(a + i)$.

- (b) With the aid of the trigonometric identities (4) in Example 3 of Sec. 10, show that the square roots obtained in part (a) can be written

$$\pm\frac{1}{\sqrt{2}}\left(\sqrt{A+a} + i\sqrt{A-a}\right).$$

(Note that this becomes the final result in Example 3, Sec. 10, when $a = \sqrt{3}$.)

6. Find the four zeros of the polynomial $z^4 + 4$, one of them being

$$z_0 = \sqrt{2}e^{i\pi/4} = 1 + i.$$

Then use those zeros to factor $z^2 + 4$ into quadratic factors with real coefficients.

Ans. $(z^2 + 2z + 2)(z^2 - 2z + 2)$.

7. Show that if c is any n th root of unity other than unity itself, then

$$1 + c + c^2 + \cdots + c^{n-1} = 0.$$

Suggestion: Use the first identity in Exercise 9, Sec. 8.

8. (a) Prove that the usual formula solves the quadratic equation

$$az^2 + bz + c = 0 \quad (a \neq 0)$$

when the coefficients a , b , and c are complex numbers. Specifically, by completing the square on the left-hand side, derive the *quadratic formula*

$$z = \frac{-b + (b^2 - 4ac)^{1/2}}{2a},$$

where both square roots are to be considered when $b^2 - 4ac \neq 0$,

(b) Use the result in part (a) to find the roots of the equation $z^2 + 2z + (1 - i) = 0$.

$$\text{Ans. (b)} \left(-1 + \frac{1}{\sqrt{2}}\right) + \frac{i}{\sqrt{2}}, \quad \left(-1 - \frac{1}{\sqrt{2}}\right) - \frac{i}{\sqrt{2}}.$$

9. Let $z = re^{i\theta}$ be a nonzero complex number and n a negative integer ($n = -1, -2, \dots$). Then define $z^{1/n}$ by means of the equation $z^{1/n} = (z^{-1})^{1/m}$ where $m = -n$. By showing that the m values of $(z^{1/m})^{-1}$ and $(z^{-1})^{1/m}$ are the same, verify that $z^{1/n} = (z^{1/m})^{-1}$. (Compare with Exercise 7, Sec. 8.)

11. REGIONS IN THE COMPLEX PLANE

In this section, we are concerned with sets of complex numbers, or points in the z plane, and their closeness to one another. Our basic tool is the concept of an ε neighborhood

$$(1) \quad |z - z_0| < \varepsilon$$

of a given point z_0 . It consists of all points z lying inside but not on a circle centered at z_0 and with a specified positive radius ε (Fig. 15). When the value of ε is understood or is immaterial in the discussion, the set (1) is often referred to as just a neighborhood. Occasionally, it is convenient to speak of a *deleted neighborhood*, or punctured disk,

$$(2) \quad 0 < |z - z_0| < \varepsilon$$

consisting of all points z in an ε neighborhood of z_0 except for the point z_0 itself.

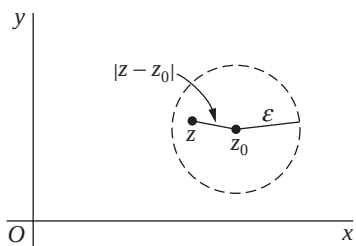


FIGURE 15

A point z_0 is said to be an *interior point* of a set S whenever there is some neighborhood of z_0 that contains only points of S ; it is called an *exterior point* of S when there exists a neighborhood of it containing no points of S . If z_0 is neither of these, it is a *boundary point* of S . A boundary point is, therefore, a point all of whose neighborhoods contain at least one point in S and at least one point not in S . The totality of all boundary points is called the *boundary* of S . The circle $|z| = 1$, for instance, is the boundary of each of the sets

$$(3) \quad |z| < 1 \quad \text{and} \quad |z| \leq 1.$$

A set is *open* if it contains none of its boundary points. It is left as an exercise to show that a set is open if and only if each of its points is an interior point. A set is *closed* if it contains all of its boundary points, and the *closure* of a set S is the closed set consisting of all points in S together with the boundary of S . Note that the first of the sets (3) is open and that the second is its closure.

Some sets are, of course, neither open nor closed. For a set to be not open, there must be a boundary point that is contained in the set; and if a set is not closed, there exists a boundary point not contained in the set. Observe that the punctured disk $0 < |z| \leq 1$ is neither open nor closed. The set of all complex numbers is, on the other hand, both open and closed since it has no boundary points.

An open set S is *connected* if each pair of points z_1 and z_2 in it can be joined by a *polygonal line*, consisting of a finite number of line segments joined end to end, that lies entirely in S . The open set $|z| < 1$ is connected. The annulus $1 < |z| < 2$ is, of course, open and it is also connected (see Fig. 16). A nonempty open set that is connected is called a *domain*. Note that any neighborhood is a domain. A domain together with some, none, or all of its boundary points is referred to as a *region*.

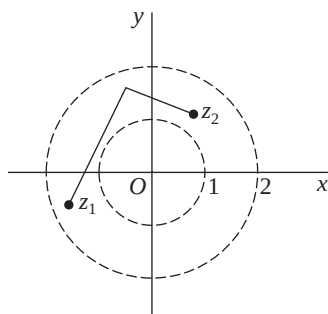


FIGURE 16

A set S is *bounded* if every point of S lies inside some circle $|z| = R$; otherwise, it is *unbounded*. Both of the sets (3) are bounded regions, and the half plane $\operatorname{Re} z \geq 0$ is unbounded.

A point z_0 is said to be an *accumulation point* of a set S if each deleted neighborhood of z_0 contains at least one point of S . It follows that if a set S is closed, then it contains each of its accumulation points. For if an accumulation point z_0 were not in S , it would be a boundary point of S ; but this contradicts the fact that a closed set contains all of its boundary points. It is left as an exercise to show that the converse is, in fact, true. Thus a set is closed if and only if it contains all of its accumulation points.

Evidently, a point z_0 is *not* an accumulation point of a set S whenever there exists some deleted neighborhood of z_0 that does not contain at least one point of S . Note that the origin is the only accumulation point of the set $z_n = i/n$ ($n = 1, 2, \dots$).

EXERCISES

1. Sketch the following sets and determine which are domains:

- (a) $|z - 2 + i| \leq 1$; (b) $|2z + 3| > 4$;
 (c) $\operatorname{Im} z > 1$; (d) $\operatorname{Im} z = 1$;
 (e) $0 \leq \arg z \leq \pi/4$ ($z \neq 0$); (f) $|z - 4| \geq |z|$.

Ans. (b), (c) are domains.

2. Which sets in Exercise 1 are neither open nor closed?

Ans. (e).

3. Which sets in Exercise 1 are bounded?

Ans. (a).

4. In each case, sketch the closure of the set:

- (a) $-\pi < \arg z < \pi$ ($z \neq 0$); (b) $|\operatorname{Re} z| < |z|$;
 (c) $\operatorname{Re}\left(\frac{1}{z}\right) \leq \frac{1}{2}$; (d) $\operatorname{Re}(z^2) > 0$.

5. Let S be the open set consisting of all points z such that $|z| < 1$ or $|z - 2| < 1$. State why S is not connected.

6. Show that a set S is open if and only if each point in S is an interior point.

7. Determine the accumulation points of each of the following sets:

- (a) $z_n = i^n$ ($n = 1, 2, \dots$); (b) $z_n = i^n/n$ ($n = 1, 2, \dots$);
 (c) $0 \leq \arg z < \pi/2$ ($z \neq 0$); (d) $z_n = (-1)^n(1 + i) \frac{n-1}{n}$ ($n = 1, 2, \dots$).

Ans. (a) None; (b) 0; (d) $\pm(1 + i)$.

8. Prove that if a set contains each of its accumulation points, then it must be a closed set.

9. Show that any point z_0 of a domain is an accumulation point of that domain.

10. Prove that a finite set of points $z_1, z_2, \dots, z_n$ cannot have any accumulation points.

CHAPTER

2

ANALYTIC FUNCTIONS

We now consider functions of a complex variable and develop a theory of differentiation for them. The main goal of the chapter is to introduce analytic functions, which play a central role in complex analysis.

12. FUNCTIONS OF A COMPLEX VARIABLE

Let S be a set of complex numbers. A *function* f defined on S is a rule that assigns to each z in S a complex number w . The number w is called the *value* of f at z and is denoted by $f(z)$; that is, $w = f(z)$. The set S is called the *domain of definition* of f .*

It must be emphasized that both a domain of definition and a rule are needed in order for a function to be well defined. When the domain of definition is not mentioned, we agree that the largest possible set is to be taken. Also, it is not always convenient to use notation that distinguishes between a given function and its values.

EXAMPLE 1. If f is defined on the set $z \neq 0$ by means of the equation $w = 1/z$, it may be referred to only as the function $w = 1/z$, or simply the function $1/z$.

Suppose that $w = u + iv$ is the value of a function f at $z = x + iy$, so that

$$u + iv = f(x + iy).$$

*Although the domain of definition is often a domain as defined in Sec. 11, it need not be.

Each of the real numbers u and v depends on the real variables x and y , and it follows that $f(z)$ can be expressed in terms of a pair of real-valued functions of the real variables x and y :

$$(1) \quad f(z) = u(x, y) + iv(x, y).$$

If the polar coordinates r and θ , instead of x and y , are used, then

$$u + iv = f(re^{i\theta})$$

where $w = u + iv$ and $z = re^{i\theta}$. In that case, we may write

$$(2) \quad f(z) = u(r, \theta) + iv(r, \theta).$$

EXAMPLE 2. If $f(z) = z^2$, then

$$f(x + iy) = (x + iy)^2 = x^2 - y^2 + i2xy.$$

Hence

$$u(x, y) = x^2 - y^2 \quad \text{and} \quad v(x, y) = 2xy.$$

When polar coordinates are used,

$$f(re^{i\theta}) = (re^{i\theta})^2 = r^2e^{i2\theta} = r^2 \cos 2\theta + ir^2 \sin 2\theta.$$

Consequently,

$$u(r, \theta) = r^2 \cos 2\theta \quad \text{and} \quad v(r, \theta) = r^2 \sin 2\theta.$$

If, in either of equations (1) and (2), the function v always has value zero, then the value of f is always real. That is, f is a *real-valued function* of a complex variable.

EXAMPLE 3. A real-valued function that is used to illustrate some important concepts later in this chapter is

$$f(z) = |z|^2 = x^2 + y^2 + i0.$$

If n is zero or a positive integer and if $a_0, a_1, a_2, \dots, a_n$ are complex constants, where $a_n \neq 0$, the function

$$P(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n$$

is a *polynomial* of degree n . Note that the sum here has a finite number of terms and that the domain of definition is the entire z plane. Quotients $P(z)/Q(z)$ of

polynomials are called *rational functions* and are defined at each point z where $Q(z) \neq 0$. Polynomials and rational functions constitute elementary, but important, classes of functions of a complex variable.

A generalization of the concept of function is a rule that assigns more than one value to a point z in the domain of definition. These *multiple-valued functions* occur in the theory of functions of a complex variable, just as they do in the case of a real variable. When multiple-valued functions are studied, usually just one of the possible values assigned to each point is taken, in a systematic manner, and a (single-valued) function is constructed from the multiple-valued function.

EXAMPLE 4. Let z denote any nonzero complex number. We know from Sec. 9 that $z^{1/2}$ has the two values

$$z^{1/2} = \pm \sqrt{r} \exp\left(i \frac{\Theta}{2}\right),$$

where $r = |z|$ and Θ ($-\pi < \Theta \leq \pi$) is the *principal value* of $\arg z$. But, if we choose only the positive value of $\pm \sqrt{r}$ and write

$$(3) \quad f(z) = \sqrt{r} \exp\left(i \frac{\Theta}{2}\right) \quad (r > 0, -\pi < \Theta \leq \pi),$$

the (single-valued) function (3) is well defined on the set of nonzero numbers in the z plane. Since zero is the only square root of zero, we also write $f(0) = 0$. The function f is then well defined on the entire plane.

EXERCISES

1. For each of the functions below, describe the domain of definition that is understood:

$$(a) f(z) = \frac{1}{z^2 + 1}; \quad (b) f(z) = \operatorname{Arg}\left(\frac{1}{z}\right);$$

$$(c) f(z) = \frac{z}{z + \bar{z}}; \quad (d) f(z) = \frac{1}{1 - |z|^2}.$$

Ans. (a) $z \neq \pm i$; (c) $\operatorname{Re} z \neq 0$.

2. Write the function $f(z) = z^3 + z + 1$ in the form $f(z) = u(x, y) + i v(x, y)$.

Ans. $f(z) = (x^3 - 3xy^2 + x + 1) + i(3x^2y - y^3 + y)$.

3. Suppose that $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$, where $z = x + iy$. Use the expressions (see Sec. 5)

$$x = \frac{z + \bar{z}}{2} \quad \text{and} \quad y = \frac{z - \bar{z}}{2i}$$

to write $f(z)$ in terms of z , and simplify the result.

Ans. $f(z) = \bar{z}^2 + 2iz$.

4. Write the function

$$f(z) = z + \frac{1}{z} \quad (z \neq 0)$$

in the form $f(z) = u(r, \theta) + i v(r, \theta)$.

$$\text{Ans. } f(z) = \left(r + \frac{1}{r}\right) \cos \theta + i \left(r - \frac{1}{r}\right) \sin \theta.$$

13. MAPPINGS

Properties of a real-valued function of a real variable are often exhibited by the graph of the function. But when $w = f(z)$, where z and w are complex, no such convenient graphical representation of the function f is available because each of the numbers z and w is located in a plane rather than on a line. One can, however, display some information about the function by indicating pairs of corresponding points $z = (x, y)$ and $w = (u, v)$. To do this, it is generally simpler to draw the z and w planes separately.

When a function f is thought of in this way, it is often referred to as a *mapping*, or transformation. The *image* of a point z in the domain of definition S is the point $w = f(z)$, and the set of images of all points in a set T that is contained in S is called the image of T . The image of the entire domain of definition S is called the *range* of f . The *inverse image* of a point w is the set of all points z in the domain of definition of f that have w as their image. The inverse image of a point may contain just one point, many points, or none at all. The last case occurs, of course, when w is not in the range of f .

Terms such as *translation*, *rotation*, and *reflection* are used to convey dominant geometric characteristics of certain mappings. In such cases, it is sometimes convenient to consider the z and w planes to be the same. For example, the mapping

$$w = z + 1 = (x + 1) + iy,$$

where $z = x + iy$, can be thought of as a translation of each point z one unit to the right. Since $i = e^{i\pi/2}$, the mapping

$$w = iz = r \exp \left[i \left(\theta + \frac{\pi}{2} \right) \right],$$

where $z = re^{i\theta}$, rotates the radius vector for each nonzero point z through a right angle about the origin in the counterclockwise direction; and the mapping

$$w = \bar{z} = x - iy$$

transforms each point $z = x + iy$ into its reflection in the real axis.

More information is usually exhibited by sketching images of curves and regions than by simply indicating images of individual points. In the following three examples, we illustrate this with the transformation $w = z^2$. We begin by finding the images of some *curves* in the z plane.

8. One interpretation of a function $w = f(z) = u(x, y) + iv(x, y)$ is that of a *vector field* in the domain of definition of f . The function assigns a vector w , with components $u(x, y)$ and $v(x, y)$, to each point z at which it is defined. Indicate graphically the vector fields represented by (a) $w = iz$; (b) $w = z/|z|$.

15. LIMITS

Let a function f be defined at all points z in some deleted neighborhood (Sec. 11) of z_0 . The statement that the *limit* of $f(z)$ as z approaches z_0 is a number w_0 , or that

$$(1) \quad \lim_{z \rightarrow z_0} f(z) = w_0,$$

means that the point $w = f(z)$ can be made arbitrarily close to w_0 if we choose the point z close enough to z_0 but distinct from it. We now express the definition of limit in a precise and usable form.

Statement (1) means that for each positive number ε , there is a positive number δ such that

$$(2) \quad |f(z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta.$$

Geometrically, this definition says that for each ε neighborhood $|w - w_0| < \varepsilon$ of w_0 , there is a deleted δ neighborhood $0 < |z - z_0| < \delta$ of z_0 such that every point z in it has an image w lying in the ε neighborhood (Fig. 23). Note that even though all points in the deleted neighborhood $0 < |z - z_0| < \delta$ are to be considered, their images need not fill up the entire neighborhood $|w - w_0| < \varepsilon$. If f has the constant value w_0 , for instance, the image of z is always the center of that neighborhood. Note, too, that once a δ has been found, it can be replaced by any smaller positive number, such as $\delta/2$.

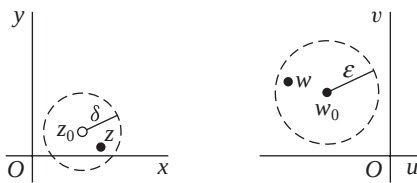


FIGURE 23

It is easy to show that when a limit of a function $f(z)$ exists at a point z_0 , it is *unique*. To do this, we suppose that

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad \text{and} \quad \lim_{z \rightarrow z_0} f(z) = w_1.$$

Then, for each positive number ε , there are positive numbers δ_0 and δ_1 such that

$$|f(z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta_0$$

and

$$|f(z) - w_1| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta_1.$$

So if $0 < |z - z_0| < \delta$, where δ is any positive number that is smaller than δ_0 and δ_1 , we find that

$$|w_1 - w_0| = |[f(z) - w_0] - [f(z) - w_1]| \leq |f(z) - w_0| + |f(z) - w_1| < \varepsilon + \varepsilon = 2\varepsilon.$$

But $|w_1 - w_0|$ is a nonnegative constant, and ε can be chosen arbitrarily small. Hence

$$w_1 - w_0 = 0, \quad \text{or} \quad w_1 = w_0.$$

Definition (2) requires that f be defined at all points in some deleted neighborhood of z_0 . Such a deleted neighborhood, of course, always exists when z_0 is an interior point of a region on which f is defined. We can extend the definition of limit to the case in which z_0 is a boundary point of the region by agreeing that the first of inequalities (2) need be satisfied by only those points z that lie in both the region *and* the deleted neighborhood.

EXAMPLE 1. Let us show that if $f(z) = i\bar{z}/2$ in the open disk $|z| < 1$, then

$$(3) \quad \lim_{z \rightarrow 1} f(z) = \frac{i}{2},$$

the point 1 being on the boundary of the domain of definition of f . Observe that when z is in the disk $|z| < 1$,

$$\left| f(z) - \frac{i}{2} \right| = \left| \frac{i\bar{z}}{2} - \frac{i}{2} \right| = \frac{|z - 1|}{2}.$$

Hence, for any such z and each positive number ε (see Fig. 24),

$$\left| f(z) - \frac{i}{2} \right| < \varepsilon \quad \text{whenever} \quad 0 < |z - 1| < 2\varepsilon.$$

Thus condition (2) is satisfied by points in the region $|z| < 1$ when δ is equal to 2ε or any smaller positive number.

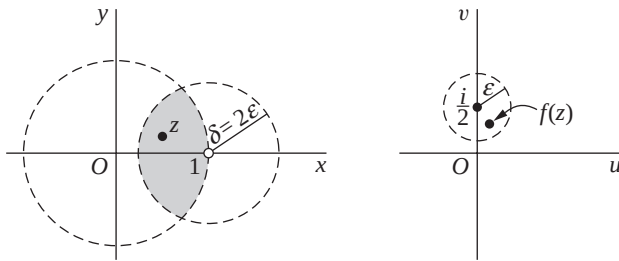


FIGURE 24

If limit (1) exists, the symbol $z \rightarrow z_0$ implies that z is allowed to approach z_0 in an arbitrary manner, not just from some particular direction. The next example emphasizes this.

EXAMPLE 2. If

$$(4) \quad f(z) = \frac{z}{\bar{z}},$$

the limit

$$(5) \quad \lim_{z \rightarrow 0} f(z)$$

does not exist. For, if it did exist, it could be found by letting the point $z = (x, y)$ approach the origin in any manner. But when $z = (x, 0)$ is a nonzero point on the real axis (Fig. 25),

$$f(z) = \frac{x + i0}{x - i0} = 1;$$

and when $z = (0, y)$ is a nonzero point on the imaginary axis,

$$f(z) = \frac{0 + iy}{0 - iy} = -1.$$

Thus, by letting z approach the origin along the real axis, we would find that the desired limit is 1. An approach along the imaginary axis would, on the other hand, yield the limit -1 . Since a limit is unique, we must conclude that limit (5) does not exist.

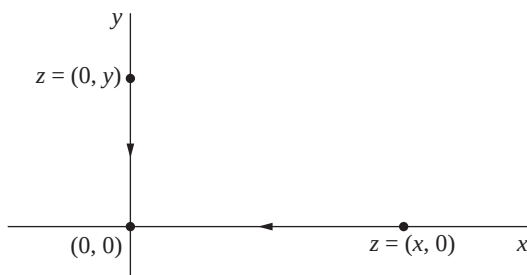


FIGURE 25

While definition (2) provides a means of testing whether a given point w_0 is a limit, it does not directly provide a method for determining that limit. Theorems on limits, presented in the next section, will enable us to actually find many limits.

16. THEOREMS ON LIMITS

We can expedite our treatment of limits by establishing a connection between limits of functions of a complex variable and limits of real-valued functions of two real variables. Since limits of the latter type are studied in calculus, we use their definition and properties freely.

Theorem 1. *Suppose that*

$$f(z) = u(x, y) + iv(x, y) \quad (z = x + iy)$$

and

$$z_0 = x_0 + iy_0, \quad w_0 = u_0 + iv_0.$$

Then

$$(1) \quad \lim_{z \rightarrow z_0} f(z) = w_0$$

if and only if

$$(2) \quad \lim_{(x,y) \rightarrow (x_0,y_0)} u(x, y) = u_0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} v(x, y) = v_0.$$

To prove the theorem, we first assume that limits (2) hold and obtain limit (1). Limits (2) tell us that for each positive number ε , there exist positive numbers δ_1 and δ_2 such that

$$(3) \quad |u - u_0| < \frac{\varepsilon}{2} \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta_1$$

and

$$(4) \quad |v - v_0| < \frac{\varepsilon}{2} \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta_2.$$

Let δ be any positive number smaller than δ_1 and δ_2 . Since

$$|(u + iv) - (u_0 + iv_0)| = |(u - u_0) + i(v - v_0)| \leq |u - u_0| + |v - v_0|$$

and

$$\sqrt{(x - x_0)^2 + (y - y_0)^2} = |(x - x_0) + i(y - y_0)| = |(x + iy) - (x_0 + iy_0)|,$$

it follows from statements (3) and (4) that

$$|(u + iv) - (u_0 + iv_0)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

whenever

$$0 < |(x + iy) - (x_0 + iy_0)| < \delta.$$

That is, limit (1) holds.

Let us now start with the assumption that limit (1) holds. With that assumption, we know that for each positive number ε , there is a positive number δ such that

$$(5) \quad |(u + iv) - (u_0 + iv_0)| < \varepsilon$$

whenever

$$(6) \quad 0 < |(x + iy) - (x_0 + iy_0)| < \delta.$$

But

$$\begin{aligned} |u - u_0| &\leq |(u - u_0) + i(v - v_0)| = |(u + iv) - (u_0 + iv_0)|, \\ |v - v_0| &\leq |(u - u_0) + i(v - v_0)| = |(u + iv) - (u_0 + iv_0)|, \end{aligned}$$

and

$$|(x + iy) - (x_0 + iy_0)| = |(x - x_0) + i(y - y_0)| = \sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

Hence it follows from inequalities (5) and (6) that

$$|u - u_0| < \varepsilon \quad \text{and} \quad |v - v_0| < \varepsilon$$

whenever

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

This establishes limits (2), and the proof of the theorem is complete.

Theorem 2. Suppose that

$$(7) \quad \lim_{z \rightarrow z_0} f(z) = w_0 \quad \text{and} \quad \lim_{z \rightarrow z_0} F(z) = W_0.$$

Then

$$(8) \quad \lim_{z \rightarrow z_0} [f(z) + F(z)] = w_0 + W_0,$$

$$(9) \quad \lim_{z \rightarrow z_0} [f(z)F(z)] = w_0 W_0;$$

and, if $W_0 \neq 0$,

$$(10) \quad \lim_{z \rightarrow z_0} \frac{f(z)}{F(z)} = \frac{w_0}{W_0}.$$

This important theorem can be proved directly by using the definition of the limit of a function of a complex variable. But, with the aid of Theorem 1, it follows almost immediately from theorems on limits of real-valued functions of two real variables.

To verify property (9), for example, we write

$$f(z) = u(x, y) + iv(x, y), \quad F(z) = U(x, y) + iV(x, y),$$

$$z_0 = x_0 + iy_0, \quad w_0 = u_0 + iv_0, \quad W_0 = U_0 + iV_0.$$

Then, according to hypotheses (7) and Theorem 1, the limits as (x, y) approaches (x_0, y_0) of the functions u , v , U , and V exist and have the values u_0 , v_0 , U_0 , and V_0 , respectively. So the real and imaginary components of the product

$$f(z)F(z) = (uU - vV) + i(vU + uV)$$

have the limits $u_0U_0 - v_0V_0$ and $v_0U_0 + u_0V_0$, respectively, as (x, y) approaches (x_0, y_0) . Hence, by Theorem 1 again, $f(z)F(z)$ has the limit

$$(u_0U_0 - v_0V_0) + i(v_0U_0 + u_0V_0)$$

as z approaches z_0 ; and this is equal to w_0W_0 . Property (9) is thus established. Corresponding verifications of properties (8) and (10) can be given.

It is easy to see from definition (2), Sec. 15, of limit that

$$\lim_{z \rightarrow z_0} c = c \quad \text{and} \quad \lim_{z \rightarrow z_0} z = z_0,$$

where z_0 and c are any complex numbers; and, by property (9) and mathematical induction, it follows that

$$\lim_{z \rightarrow z_0} z^n = z_0^n \quad (n = 1, 2, \dots).$$

So, in view of properties (8) and (9), the limit of a polynomial

$$P(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n$$

as z approaches a point z_0 is the value of the polynomial at that point:

$$(11) \quad \lim_{z \rightarrow z_0} P(z) = P(z_0).$$

17. LIMITS INVOLVING THE POINT AT INFINITY

It is sometimes convenient to include with the complex plane the *point at infinity*, denoted by ∞ , and to use limits involving it. The complex plane together with this point is called the *extended complex plane*. To visualize the point at infinity, one can think of the complex plane as passing through the equator of a unit sphere centered at the origin (Fig. 26). To each point z in the plane there corresponds exactly one point P on the surface of the sphere. The point P is the point where the line through z and the north pole N intersects the sphere. In like manner, to each point P on the surface of the sphere, other than the north pole N , there corresponds exactly one

18. CONTINUITY

A function f is *continuous* at a point z_0 if all three of the following conditions are satisfied:

- (1) $\lim_{z \rightarrow z_0} f(z)$ exists,
- (2) $f(z_0)$ exists,
- (3) $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Observe that statement (3) actually contains statements (1) and (2), since the existence of the quantity on each side of the equation there is needed. Statement (3) says, of course, that for each positive number ε , there is a positive number δ such that

$$(4) \quad |f(z) - f(z_0)| < \varepsilon \quad \text{whenever} \quad |z - z_0| < \delta.$$

A function of a complex variable is said to be continuous in a region R if it is continuous at each point in R .

If two functions are continuous at a point, their sum and product are also continuous at that point; their quotient is continuous at any such point if the denominator is not zero there. These observations are direct consequences of Theorem 2, Sec. 16. Note, too, that a polynomial is continuous in the entire plane because of limit (11) in Sec. 16.

We turn now to two expected properties of continuous functions whose verifications are not so immediate. Our proofs depend on definition (4) of continuity, and we present the results as theorems.

Theorem 1. *A composition of continuous functions is itself continuous.*

A precise statement of this theorem is contained in the proof to follow. We let $w = f(z)$ be a function that is defined for all z in a neighborhood $|z - z_0| < \delta$ of a point z_0 , and we let $W = g(w)$ be a function whose domain of definition contains the image (Sec. 13) of that neighborhood under f . The composition $W = g[f(z)]$ is, then, defined for all z in the neighborhood $|z - z_0| < \delta$. Suppose now that f is continuous at z_0 and that g is continuous at the point $f(z_0)$ in the w plane. In view of the continuity of g at $f(z_0)$, there is, for each positive number ε , a positive number γ such that

$$|g[f(z)] - g[f(z_0)]| < \varepsilon \quad \text{whenever} \quad |f(z) - f(z_0)| < \gamma.$$

(See Fig. 27.) But the continuity of f at z_0 ensures that the neighborhood $|z - z_0| < \delta$ can be made small enough that the second of these inequalities holds. The continuity of the composition $g[f(z)]$ is, therefore, established.

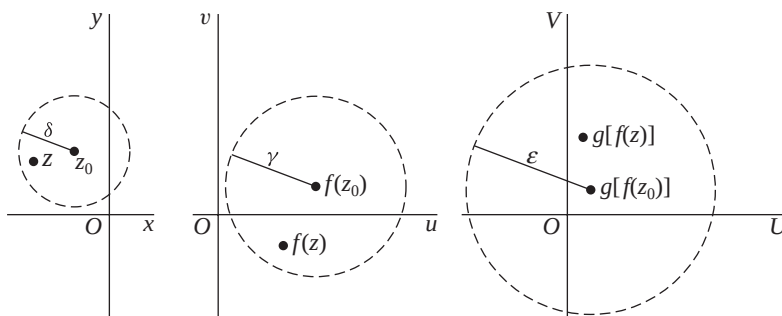


FIGURE 27

Theorem 2. If a function $f(z)$ is continuous and nonzero at a point z_0 , then $f(z) \neq 0$ throughout some neighborhood of that point.

Assuming that $f(z)$ is, in fact, continuous and nonzero at z_0 , we can prove Theorem 2 by assigning the positive value $|f(z_0)|/2$ to the number ε in statement (4). This tells us that there is a positive number δ such that

$$|f(z) - f(z_0)| < \frac{|f(z_0)|}{2} \quad \text{whenever} \quad |z - z_0| < \delta.$$

So if there is a point z in the neighborhood $|z - z_0| < \delta$ at which $f(z) = 0$, we have the contradiction

$$|f(z_0)| < \frac{|f(z_0)|}{2},$$

and the theorem is proved.

The continuity of a function

$$(5) \quad f(z) = u(x, y) + iv(x, y)$$

is closely related to the continuity of its component functions $u(x, y)$ and $v(x, y)$. We note, for instance, how it follows from Theorem 1 in Sec. 16 that the function (5) is continuous at a point $z_0 = (x_0, y_0)$ if and only if its component functions are continuous there. Our proof of the next theorem illustrates the use of this statement. The theorem is extremely important and will be used often in later chapters, especially in applications. Before stating the theorem, we recall from Sec. 11 that a region R is closed if it contains all of its boundary points and that it is bounded if it lies inside some circle centered at the origin.

Theorem 3. If a function f is continuous throughout a region R that is both closed and bounded, there exists a nonnegative real number M such that

$$(6) \quad |f(z)| \leq M \quad \text{for all points } z \text{ in } R,$$

where equality holds for at least one such z .

To prove this, we assume that the function f in equation (5) is continuous and note how it follows that the function

$$\sqrt{[u(x, y)]^2 + [v(x, y)]^2}$$

is continuous throughout R and thus reaches a maximum value M somewhere in R .^{*} Inequality (6) thus holds, and we say that f is *bounded on R* .

EXERCISES

1. Use definition (2), Sec. 15, of limit to prove that

$$(a) \lim_{z \rightarrow z_0} \operatorname{Re} z = \operatorname{Re} z_0; \quad (b) \lim_{z \rightarrow z_0} \bar{z} = \bar{z}_0; \quad (c) \lim_{z \rightarrow 0} \frac{\bar{z}^2}{z} = 0.$$

2. Let a , b , and c denote complex constants. Then use definition (2), Sec. 15, of limit to show that

$$(a) \lim_{z \rightarrow z_0} (az + b) = az_0 + b; \quad (b) \lim_{z \rightarrow z_0} (z^2 + c) = z_0^2 + c;$$

$$(c) \lim_{z \rightarrow 1-i} [x + i(2x + y)] = 1 + i \quad (z = x + iy).$$

3. Let n be a positive integer and let $P(z)$ and $Q(z)$ be polynomials, where $Q(z_0) \neq 0$. Use Theorem 2 in Sec. 16, as well as limits appearing in that section, to find

$$(a) \lim_{z \rightarrow z_0} \frac{1}{z^n} \quad (z_0 \neq 0); \quad (b) \lim_{z \rightarrow i} \frac{iz^3 - 1}{z + i}; \quad (c) \lim_{z \rightarrow z_0} \frac{P(z)}{Q(z)}.$$

$$\text{Ans. } (a) 1/z_0^n; \quad (b) 0; \quad (c) P(z_0)/Q(z_0).$$

4. Use mathematical induction and property (9), Sec. 16, of limits to show that

$$\lim_{z \rightarrow z_0} z^n = z_0^n$$

when n is a positive integer ($n = 1, 2, \dots$).

5. Show that the limit of the function

$$f(z) = \left(\frac{z}{\bar{z}}\right)^2$$

as z tends to 0 does not exist. Do this by letting nonzero points $z = (x, 0)$ and $z = (x, x)$ approach the origin. [Note that it is not sufficient to simply consider points $z = (x, 0)$ and $z = (0, y)$, as it was in Example 2, Sec. 15.]

6. Prove statement (8) in Theorem 2 of Sec. 16 using

- (a) Theorem 1 in Sec. 16 and properties of limits of real-valued functions of two real variables;
(b) definition (2), Sec. 15, of limit.

^{*}See, for instance, A. E. Taylor and W. R. Mann, "Advanced Calculus," 3d ed., pp. 125–126 and p. 529, 1983.

7. Use definition (2), Sec. 15, of limit to prove that

$$\text{if } \lim_{z \rightarrow z_0} f(z) = w_0, \quad \text{then } \lim_{z \rightarrow z_0} |f(z)| = |w_0|.$$

Suggestion: Observe how the first of inequalities (9), Sec. 4, enables one to write

$$||f(z)| - |w_0|| \leq |f(z) - w_0|.$$

8. Write $\Delta z = z - z_0$ and show that

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad \text{if and only if} \quad \lim_{\Delta z \rightarrow 0} f(z_0 + \Delta z) = w_0.$$

9. Show that

$$\lim_{z \rightarrow z_0} f(z)g(z) = 0 \quad \text{if} \quad \lim_{z \rightarrow z_0} f(z) = 0$$

and if there exists a positive number M such that $|g(z)| \leq M$ for all z in some neighborhood of z_0 .

10. Use the theorem in Sec. 17 to show that

$$(a) \lim_{z \rightarrow \infty} \frac{4z^2}{(z-1)^2} = 4; \quad (b) \lim_{z \rightarrow 1} \frac{1}{(z-1)^3} = \infty; \quad (c) \lim_{z \rightarrow \infty} \frac{z^2+1}{z-1} = \infty.$$

11. With the aid of the theorem in Sec. 17, show that when

$$T(z) = \frac{az+b}{cz+d} \quad (ad-bc \neq 0),$$

$$(a) \lim_{z \rightarrow \infty} T(z) = \infty \quad \text{if } c = 0;$$

$$(b) \lim_{z \rightarrow \infty} T(z) = \frac{a}{c} \quad \text{and} \quad \lim_{z \rightarrow -d/c} T(z) = \infty \quad \text{if } c \neq 0.$$

12. State why limits involving the point at infinity are unique.

13. Show that a set S is unbounded (Sec. 11) if and only if every neighborhood of the point at infinity contains at least one point in S .

19. DERIVATIVES

Let f be a function whose domain of definition contains a neighborhood $|z - z_0| < \varepsilon$ of a point z_0 . The *derivative* of f at z_0 is the limit

$$(1) \quad f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

and the function f is said to be *differentiable* at z_0 when $f'(z_0)$ exists.

By expressing the variable z in definition (1) in terms of the new complex variable

$$\Delta z = z - z_0 \quad (z \neq z_0),$$

one can write that definition as

$$(2) \quad f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

Because f is defined throughout a neighborhood of z_0 , the number $f(z_0 + \Delta z)$ is always defined for $|\Delta z|$ sufficiently small (Fig. 28).

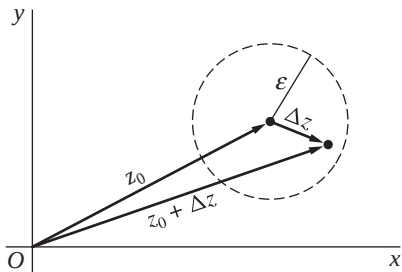


FIGURE 28

When taking form (2) of the definition of derivative, we often drop the subscript on z_0 and introduce the number

$$\Delta w = f(z + \Delta z) - f(z),$$

which denotes the change in the value $w = f(z)$ of f corresponding to a change Δz in the point at which f is evaluated. Then, if we write dw/dz for $f'(z)$, equation (2) becomes

$$(3) \quad \frac{dw}{dz} = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z}.$$

EXAMPLE 1. Suppose that $f(z) = z^2$. At any point z ,

$$\lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z$$

since $2z + \Delta z$ is a polynomial in Δz . Hence $dw/dz = 2z$, or $f'(z) = 2z$.

EXAMPLE 2. If $f(z) = \bar{z}$, then

$$(4) \quad \frac{\Delta w}{\Delta z} = \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \frac{\bar{z} + \overline{\Delta z} - \bar{z}}{\Delta z} = \frac{\overline{\Delta z}}{\Delta z}.$$

If the limit of $\Delta w/\Delta z$ exists, it can be found by letting the point $\Delta z = (\Delta x, \Delta y)$ approach the origin $(0, 0)$ in the Δz plane in any manner. In particular, as Δz approaches $(0, 0)$ horizontally through the points $(\Delta x, 0)$ on the real axis (Fig. 29),

$$\overline{\Delta z} = \overline{\Delta x + i0} = \Delta x - i0 = \Delta x + i0 = \Delta z.$$

In that case, expression (4) tells us that

$$\frac{\Delta w}{\Delta z} = \frac{\Delta z}{\Delta z} = 1.$$

Hence if the limit of $\Delta w/\Delta z$ exists, its value must be unity. However, when Δz approaches $(0, 0)$ vertically through the points $(0, \Delta y)$ on the imaginary axis, so that

$$\overline{\Delta z} = \overline{0 + i\Delta y} = 0 - i\Delta y = -(0 + i\Delta y) = -\Delta z,$$

we find from expression (4) that

$$\frac{\Delta w}{\Delta z} = \frac{-\Delta z}{\Delta z} = -1.$$

Hence the limit must be -1 if it exists. Since limits are unique (Sec. 15), it follows that dw/dz does not exist anywhere.

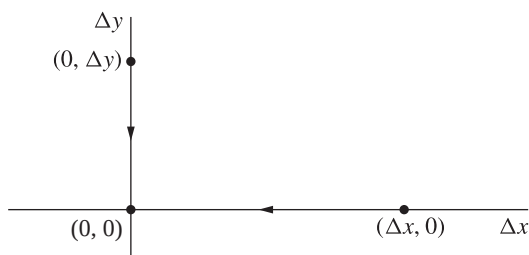


FIGURE 29

EXAMPLE 3. Consider the real-valued function $f(z) = |z|^2$. Here

$$(5) \quad \frac{\Delta w}{\Delta z} = \frac{|z + \Delta z|^2 - |z|^2}{\Delta z} = \frac{(z + \Delta z)(\bar{z} + \overline{\Delta z}) - z\bar{z}}{\Delta z} = \bar{z} + \overline{\Delta z} + z \frac{\overline{\Delta z}}{\Delta z}.$$

Proceeding as in Example 2, where horizontal and vertical approaches of Δz toward the origin gave us

$$\overline{\Delta z} = \Delta z \quad \text{and} \quad \overline{\Delta z} = -\Delta z,$$

respectively, we have the expressions

$$\frac{\Delta w}{\Delta z} = \bar{z} + \Delta z + z \quad \text{when} \quad \Delta z = (\Delta x, 0)$$

and

$$\frac{\Delta w}{\Delta z} = \bar{z} - \Delta z - z \quad \text{when} \quad \Delta z = (0, \Delta y).$$

Hence if the limit of $\Delta w/\Delta z$ exists as Δz tends to zero, the uniqueness of limits, used in Example 2, tells us that

$$\bar{z} + z = \bar{z} - z,$$

or $z = 0$. Evidently, then dw/dz cannot exist when $z \neq 0$.

To show that dw/dz does, in fact, exist at $z = 0$, we need only observe that expression (5) reduces to

$$\frac{\Delta w}{\Delta z} = \overline{\Delta z}$$

when $z = 0$. We conclude, therefore, that dw/dz exists *only* at $z = 0$, its value there being 0.

Example 3 shows that a function $f(z) = u(x, y) + iv(x, y)$ can be differentiable at a point $z = (x, y)$ but nowhere else in any neighborhood of that point. Since

$$(6) \quad u(x, y) = x^2 + y^2 \quad \text{and} \quad v(x, y) = 0$$

when $f(z) = |z|^2$, it also shows that the real and imaginary components of a function of a complex variable can have continuous partial derivatives of all orders at a point $z = (x, y)$ and yet the function may not be differentiable there.

The function $f(z) = |z|^2$ is continuous at each point in the plane since its components (6) are continuous at each point. So the continuity of a function at a point does not imply the existence of a derivative there. It is, however, true that *the existence of the derivative of a function at a point implies the continuity of the function at that point*. To see this, we assume that $f'(z_0)$ exists and write

$$\lim_{z \rightarrow z_0} [f(z) - f(z_0)] = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \lim_{z \rightarrow z_0} (z - z_0) = f'(z_0) \cdot 0 = 0,$$

from which it follows that

$$\lim_{z \rightarrow z_0} f(z) = f(z_0).$$

This is the statement of continuity of f at z_0 (Sec. 18).

Geometric interpretations of derivatives of functions of a complex variable are not as immediate as they are for derivatives of functions of a real variable. We defer the development of such interpretations until Chap. 9.

20. DIFFERENTIATION FORMULAS

The definition of derivative in Sec. 19 is identical in form to that of the derivative of a real-valued function of a real variable. In fact, the basic differentiation formulas given below can be derived from the definition in Sec. 19 by essentially the same steps as the ones used in calculus. In these formulas, the derivative of a function f at a point z is denoted by either

$$\frac{d}{dz}f(z) \quad \text{or} \quad f'(z),$$

depending on which notation is more convenient.

Let c be a complex constant, and let f be a function whose derivative exists at a point z . It is easy to show that

$$(1) \quad \frac{d}{dz}c = 0, \quad \frac{d}{dz}z = 1, \quad \frac{d}{dz}[cf(z)] = cf'(z).$$

Also, if n is a positive integer,

$$(2) \quad \frac{d}{dz}z^n = nz^{n-1}.$$

This formula remains valid when n is a negative integer, provided that $z \neq 0$.

If the derivatives of two functions f and g exist at a point z , then

$$(3) \quad \frac{d}{dz}[f(z) + g(z)] = f'(z) + g'(z),$$

$$(4) \quad \frac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z);$$

and, when $g(z) \neq 0$,

$$(5) \quad \frac{d}{dz} \left[\frac{f(z)}{g(z)} \right] = \frac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2}.$$

Let us derive formula (4). To do this, we write the following expression for the change in the product $w = f(z)g(z)$:

$$\begin{aligned} \Delta w &= f(z + \Delta z)g(z + \Delta z) - f(z)g(z) \\ &= f(z)[g(z + \Delta z) - g(z)] + [f(z + \Delta z) - f(z)]g(z + \Delta z). \end{aligned}$$

Thus

$$\frac{\Delta w}{\Delta z} = f(z) \frac{g(z + \Delta z) - g(z)}{\Delta z} + \frac{f(z + \Delta z) - f(z)}{\Delta z} g(z + \Delta z);$$

and, letting Δz tend to zero, we arrive at the desired formula for the derivative of $f(z)g(z)$. Here we have used the fact that g is continuous at the point z , since

$g'(z)$ exists; thus $g(z + \Delta z)$ tends to $g(z)$ as Δz tends to zero (see Exercise 8, Sec. 18).

There is also a chain rule for differentiating composite functions. Suppose that f has a derivative at z_0 and that g has a derivative at the point $f(z_0)$. Then the function $F(z) = g[f(z)]$ has a derivative at z_0 , and

$$(6) \quad F'(z_0) = g'[f(z_0)]f'(z_0).$$

If we write $w = f(z)$ and $W = g(w)$, so that $W = F(z)$, the chain rule becomes

$$\frac{dW}{dz} = \frac{dW}{dw} \frac{dw}{dz}.$$

EXAMPLE. To find the derivative of $(2z^2 + i)^5$, write $w = 2z^2 + i$ and $W = w^5$. Then

$$\frac{d}{dz}(2z^2 + i)^5 = 5w^4 4z = 20z(2z^2 + i)^4.$$

To start the derivation of formula (6), choose a specific point z_0 at which $f'(z_0)$ exists. Write $w_0 = f(z_0)$ and also assume that $g'(w_0)$ exists. There is, then, some ε neighborhood $|w - w_0| < \varepsilon$ of w_0 such that for all points w in that neighborhood, we can define a function Φ having the values $\Phi(w_0) = 0$ and

$$(7) \quad \Phi(w) = \frac{g(w) - g(w_0)}{w - w_0} - g'(w_0) \quad \text{when } w \neq w_0.$$

Note that in view of the definition of derivative,

$$(8) \quad \lim_{w \rightarrow w_0} \Phi(w) = 0.$$

Hence Φ is continuous at w_0 .

Now expression (7) can be put in the form

$$(9) \quad g(w) - g(w_0) = [g'(w_0) + \Phi(w)](w - w_0) \quad (|w - w_0| < \varepsilon),$$

which is valid even when $w = w_0$; and since $f'(z_0)$ exists and f is therefore continuous at z_0 , we can choose a positive number δ such that the point $f(z)$ lies in the ε neighborhood $|w - w_0| < \varepsilon$ of w_0 if z lies in the δ neighborhood $|z - z_0| < \delta$ of z_0 . Thus it is legitimate to replace the variable w in equation (9) by $f(z)$ when z is any point in the neighborhood $|z - z_0| < \delta$. With that substitution, and with $w_0 = f(z_0)$, equation (9) becomes

$$(10) \quad \frac{g[f(z)] - g[f(z_0)]}{z - z_0} = \{g'[f(z_0)] + \Phi[f(z)]\} \frac{f(z) - f(z_0)}{z - z_0} \quad (0 < |z - z_0| < \delta),$$

where we must stipulate that $z \neq z_0$ so that we are not dividing by zero. As already noted, f is continuous at z_0 and Φ is continuous at the point $w_0 = f(z_0)$. Hence the composition $\Phi[f(z)]$ is continuous at z_0 ; and since $\Phi(w_0) = 0$,

$$\lim_{z \rightarrow z_0} \Phi[f(z)] = 0.$$

So equation (10) becomes equation (6) in the limit as z approaches z_0 .

EXERCISES

1. Use results in Sec. 20 to find $f'(z)$ when

$$\begin{aligned} (a) \ f(z) &= 3z^2 - 2z + 4; & (b) \ f(z) &= (1 - 4z^2)^3; \\ (c) \ f(z) &= \frac{z-1}{2z+1} \ (z \neq -1/2); & (d) \ f(z) &= \frac{(1+z^2)^4}{z^2} \ (z \neq 0). \end{aligned}$$

2. Using results in Sec. 20, show that

(a) a polynomial

$$P(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n \quad (a_n \neq 0)$$

of degree n ($n \geq 1$) is differentiable everywhere, with derivative

$$P'(z) = a_1 + 2a_2z + \cdots + na_nz^{n-1};$$

(b) the coefficients in the polynomial $P(z)$ in part (a) can be written

$$a_0 = P(0), \quad a_1 = \frac{P'(0)}{1!}, \quad a_2 = \frac{P''(0)}{2!}, \quad \dots, \quad a_n = \frac{P^{(n)}(0)}{n!}.$$

3. Apply definition (3), Sec. 19, of derivative to give a direct proof that

$$\frac{dw}{dz} = -\frac{1}{z^2} \quad \text{when} \quad w = \frac{1}{z} \quad (z \neq 0).$$

4. Suppose that $f(z_0) = g(z_0) = 0$ and that $f'(z_0)$ and $g'(z_0)$ exist, where $g'(z_0) \neq 0$. Use definition (1), Sec. 19, of derivative to show that

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

5. Derive formula (3), Sec. 20, for the derivative of the sum of two functions.

6. Derive expression (2), Sec. 20, for the derivative of z^n when n is a positive integer by using

(a) mathematical induction and formula (4), Sec. 20, for the derivative of the product of two functions;

(b) definition (3), Sec. 19, of derivative and the binomial formula (Sec. 3).

7. Prove that expression (2), Sec. 20, for the derivative of z^n remains valid when n is a negative integer ($n = -1, -2, \dots$), provided that $z \neq 0$.

Suggestion: Write $m = -n$ and use the formula for the derivative of a quotient of two functions.

8. Use the method in Example 2, Sec. 19, to show that $f'(z)$ does not exist at any point z when
 (a) $f(z) = \operatorname{Re} z$; (b) $f(z) = \operatorname{Im} z$.
9. Let f denote the function whose values are

$$f(z) = \begin{cases} \bar{z}^2/z & \text{when } z \neq 0, \\ 0 & \text{when } z = 0. \end{cases}$$

Show that if $z = 0$, then $\Delta w/\Delta z = 1$ at each nonzero point on the real and imaginary axes in the Δz , or $\Delta x \Delta y$, plane. Then show that $\Delta w/\Delta z = -1$ at each nonzero point $(\Delta x, \Delta x)$ on the line $\Delta y = \Delta x$ in that plane. Conclude from these observations that $f'(0)$ does not exist. Note that to obtain this result, it is not sufficient to consider only horizontal and vertical approaches to the origin in the Δz plane. (Compare with Example 2, Sec. 19.)

21. CAUCHY-RIEMANN EQUATIONS

In this section, we obtain a pair of equations that the first-order partial derivatives of the component functions u and v of a function

$$(1) \quad f(z) = u(x, y) + i v(x, y)$$

must satisfy at a point $z_0 = (x_0, y_0)$ when the derivative of f exists there. We also show how to express $f'(z_0)$ in terms of those partial derivatives.

We start by writing

$$z_0 = x_0 + i y_0, \quad \Delta z = \Delta x + i \Delta y,$$

and

$$\begin{aligned} \Delta w &= f(z_0 + \Delta z) - f(z_0) \\ &= [u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0)] + i[v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0)]. \end{aligned}$$

Assuming that the derivative

$$(2) \quad f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z}$$

exists, we know from Theorem 1 in Sec. 16 that

$$(3) \quad f'(z_0) = \lim_{(\Delta x, \Delta y) \rightarrow (0,0)} \left(\operatorname{Re} \frac{\Delta w}{\Delta z} \right) + i \lim_{(\Delta x, \Delta y) \rightarrow (0,0)} \left(\operatorname{Im} \frac{\Delta w}{\Delta z} \right).$$